Invariance and Its Limits in Information Theory, Compression, and Finance

Formal Metrology Working Group

Abstract—This paper extends the series to three further domains, combined for efficiency. We prove: (1) Shannon entropy is exactly invariant under relabeling of outcomes but strictly decreases under lossy binning (a direct information-theoretic statement of the data-processing inequality), (2) practical fixed-rate uniform quantization approaches, but never reaches, the Shannon rate-distortion bound for a Gaussian source, converging to an empirically constant rate penalty — the compression-domain analogue of the ADC quantization results earlier in this series, and (3) the Black-Scholes call option price is exactly homogeneous of degree one in the spot price and strike price jointly, the finance-domain analogue of the power-law homogeneity proved earlier for physics. All three are verified numerically.

1 Entropy: Invariant Under Relabeling, Not Under Binning

Shannon entropy $H(X) = -\sum_i p_i \log_2 p_i$ depends only on the *probabilities* assigned to outcomes, not on how those outcomes are labeled or ordered.

Proposition 1. *For any permutation π of outcome labels, $H(X_\pi) = H(X)$ exactly. For any deterministic, non-injective merging of outcomes (binning), $H(\text{binned}) \leq H(X)$, with equality only if the merge combines zero-probability outcomes.*

Proof. The first claim is immediate: H is a sum over the multiset of probabilities $\{p_i\}$, unchanged by which label each p_i is attached to. The second is the data-processing inequality applied to a deterministic function of X: any deterministic mapping f satisfies $H(f(X)) \leq H(X)$, since $f(X)$ is a coarsening — some distinct outcomes become indistinguishable, which can only reduce (never increase) the information needed to describe the result. □

1.1 Worked numerical verification

For outcome probabilities $\{0.4, 0.25, 0.2, 0.1, 0.05\}$, $H = 2.041446$ bits. Randomly permuting the labels reproduces $H = 2.041446$ bits exactly. Merging the two largest outcomes into one bin (a lossy, coarser labeling) drops entropy to $H_{\text{merged}} = 1.416642$ bits — confirming $H_{\text{merged}} \leq H$, as the data-processing inequality requires.

2 Compression: Approaching, Never Reaching, the Rate-Distortion Bound

For a Gaussian source with variance σ^2, Shannon's rate-distortion theorem gives the minimum bit rate needed to achieve mean-squared distortion D:

$$R(D) = \tfrac{1}{2} \log_2\left(\frac{\sigma^2}{D}\right). \tag{1}$$

This is the theoretical floor; practical fixed-rate uniform quantizers (as used, in transform-coefficient form, by JPEG and MP3-style codecs) cannot reach it exactly.

2.1 Worked numerical verification

Quantizing $500{,}000$ samples from $\mathcal{N}(0,1)$ with a uniform quantizer over $[-4, 4]$ at several bit depths, and comparing the achieved distortion D to the rate $R(D)$ the Shannon bound would permit for that same D:

TABLE 1
Uniform quantization vs. the Shannon rate-distortion bound

Bits used	Distortion D	$R(D)$ bound	Gap (bits)
2	0.330442	0.799	1.201
4	0.020820	2.793	1.207
6	0.001308	4.789	1.211
8	0.000087	6.743	1.257

At every bit depth tested, the uniform quantizer requires roughly 1.2 more bits than the Shannon-optimal rate for the same distortion — and this gap stays nearly constant as bit depth increases, rather than shrinking to zero. This matches the well-established theoretical result that fixed-rate uniform scalar quantization incurs an asymptotically constant rate penalty relative to $R(D)$ (the classical entropy-coded version of this penalty is the well-known 1.53 dB, or ≈ 0.25 bit, gap; the larger ≈ 1.2-bit gap confirmed here reflects the additional cost of *fixed*-length, non-entropy-coded symbols, consistent with why real codecs like JPEG apply entropy coding, e.g. Huffman coding, *after* quantization rather than relying on the raw quantizer alone).

3 Finance: Homogeneity of the Black-Scholes Price

The Black-Scholes call price is

$$C(S, K, T, r, \sigma) = S\,\Phi(d_1) - Ke^{-rT}\,\Phi(d_2), \qquad d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma}{\sigma\sqrt{T}} \tag{2}$$

Proposition 2. *For any $c > 0$, $C(cS, cK, T, r, \sigma) = c\,C(S, K, T, r, \sigma)$.*

Proof. d_1 depends on S, K only through $\ln(S/K)$, which is unchanged by a common positive scaling: $\ln(cS/cK) = \ln(S/K)$. Hence d_1, d_2 (and therefore $\Phi(d_1), \Phi(d_2)$) are unchanged. Then $C(cS, cK, \ldots) = cS\,\Phi(d_1) - cKe^{-rT}\Phi(d_2) = c\big[S\Phi(d_1) - Ke^{-rT}\Phi(d_2)\big] = c\,C(S, K, \ldots)$. $\square$

3.1 Worked numerical verification

With $S = 100$, $K = 95$, $T = 1$, $r = 0.03$, $\sigma = 0.25$: $C = 13.961178$. Scaling both S and K by $c \in \{2.0,\ 0.5,\ 10.0,\ 0.1\}$ reproduces $c \cdot C$ exactly in every case (e.g., $c = 10$: $C(1000, 950, \ldots) = 139.611783 = 10 \times 13.961178$). This is the finance-domain instance of the same power-law/linear homogeneity proved for physical power laws earlier in this series: an option's price responds proportionally to a common rescaling of the price scale in which both the underlying and the strike are quoted (e.g., switching currency, or a stock split combined with a proportional strike adjustment), because only the *ratio* S/K, not the absolute scale, enters the pricing formula.

4 Conclusion

Three more domains, three more instances of the same distinction that has organized this entire series:

- Entropy is exactly invariant to relabeling but strictly decreases under lossy coarsening — confirmed exactly (permutation) and directionally (binning) in Section I.
- Fixed-rate quantization approaches but never reaches the Shannon-optimal rate-distortion tradeoff, converging to a measurable, roughly constant gap — confirmed at ≈ 1.2 bits across four bit depths in Section II.
- Black-Scholes option pricing is exactly homogeneous of degree one in (S, K) jointly — confirmed exactly across a $100\times$ range of scale factors in Section III.

As throughout: whether a transformation preserves structure is never a matter of intuition or rhetorical framing — it is a specific, provable claim, and this series has now checked that claim across nine domains.

References

1) C. E. Shannon, "A Mathematical Theory of Communication," Bell System Technical Journal, 1948.
2) T. Berger, *Rate Distortion Theory*, Prentice-Hall, 1971.
3) F. Black and M. Scholes, "The Pricing of Options and Corporate Liabilities," Journal of Political Economy, 1973.